Monetary Policy, the Financial Cycle and Ultra-Low Interest Rates

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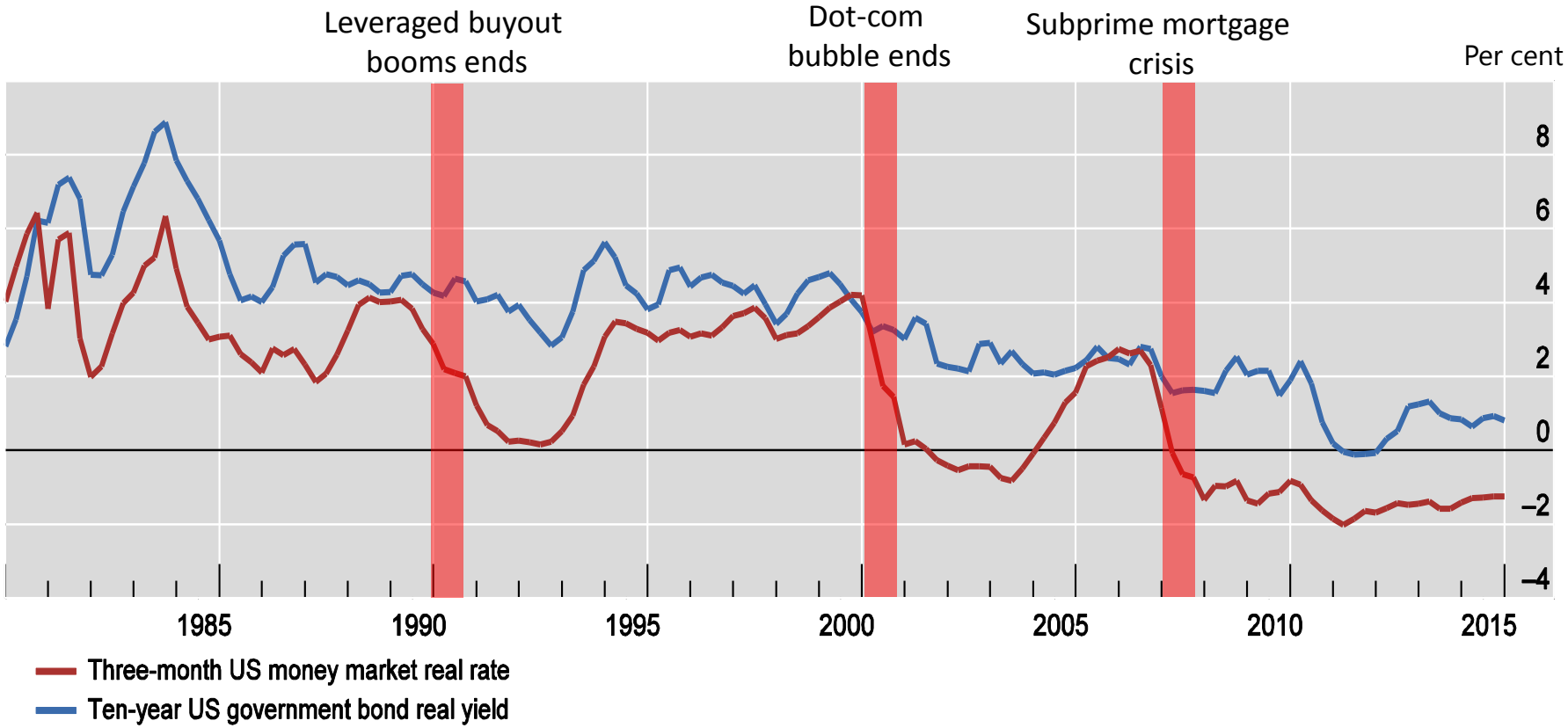
Why Are Real Rates So Low?

The long-term decline in real interest rates



The Standard View…

Exogenous decline in the natural rate

- Saving-investment fundamentals
  - Saving glut, safe asset shortage, demographics etc.
- Implications (of low natural rate)
  - ZLB binds more often (eg. secular stagnation)
  - Need for higher inflation targets
- Inferred from inflation behaviour

…Is Incomplete

But…

- Is inflation a sufficient indicator of sustainability?
  - Failure of Phillip's Curve
  - Financial imbalance build up without inflation

- Financial cycles have very persistent impacts
  - Effects of financial crises permanent
  - Monetary policy may not be neutral

What We Do

Incorporate the role of financial cycle

- How does finance affect *trends* and, hence, our assessments of the output gap and natural rate?
- What is the contribution of monetary policy?

Takeaways

- Natural rate higher and decline by less when account for financial cycle
- Monetary policy has long-run effects
- Policy should *systematically* respond to financial cycle and focus on long horizons

Characterizing The Financial Cycle

Key Ideas

- Financial cycle can be anchored to two long-run relationships that pin down sustainable credit-GDP ratio
  - Leverage (credit-to-assets):

$$lev_t = (cr_t^r - y_t^r) - \beta_{lev} p^r{}_{A,t}$$

  - Debt service ratio (DSR)

$$dsr_t = (cr_t^r - y_t^r) + \beta_{dsr} i_{L,t}$$

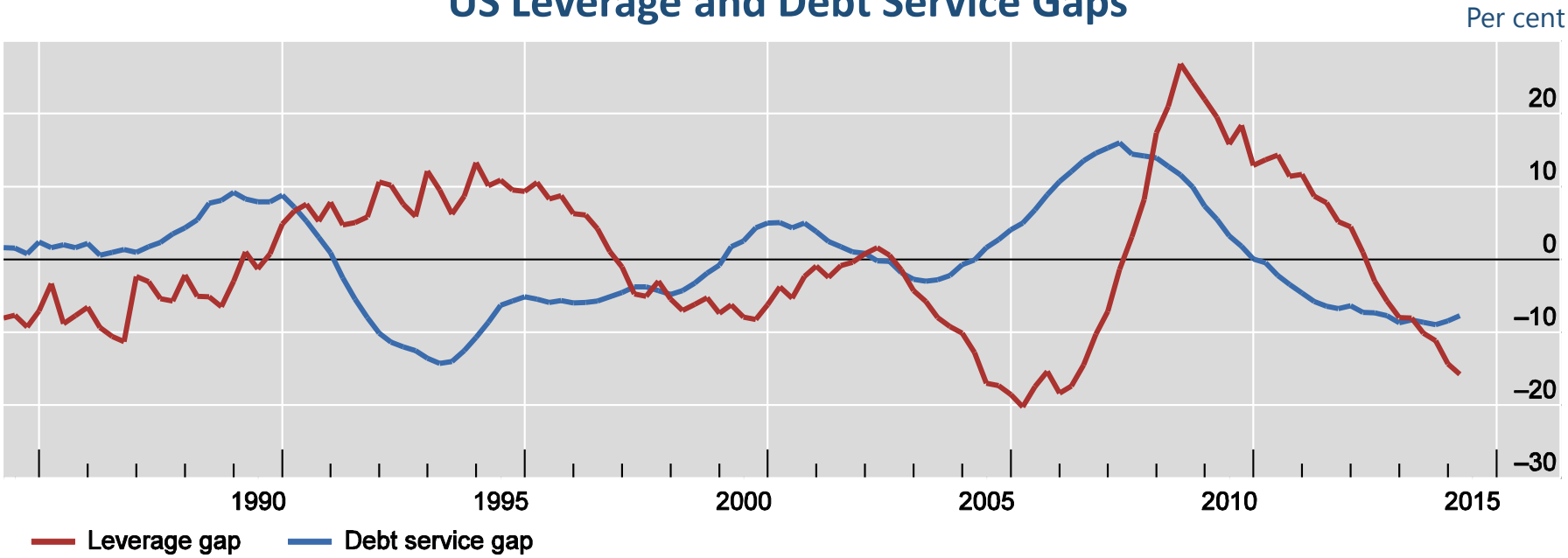
where cr_t^r is real credit, y_t^r is real output, $p^r{}_{A,t}$ is real asset price, $i_{L,t}$ is the nominal average lending rate on the stock of credit

- Credit-to-GDP, real asset prices, and nominal lending rate proportional in the long-run

Characterizing The Financial Cycle

- Financial imbalances: Deviations from long-run average
  - Leverage gap: $\widetilde{lev}_t = lev_t - \overline{lev}$
  - Debt service gap: $\widetilde{dsr}_t = dsr_t - \overline{dsr}$

US Leverage and Debt Service Gaps



Source: Juselius et al. (2016)

Output - Financial Cycle Nexus

Output Effects

Embed in a VAR with output, inflation, policy rate

$$\begin{pmatrix} \Delta cr^r \\ \Delta e_P^r \\ \Delta e_O^r \\ \Delta p_A^r \\ \Delta \pi \\ \Delta i_L \\ \Delta i \end{pmatrix}_t = \gamma_0 + \alpha \underbrace{\begin{pmatrix} \widetilde{lev} \\ \widetilde{dsr} \\ \widetilde{spr} \end{pmatrix}_{t-1}}_{\text{Steady-state deviations}} + \sum_{j=1}^{3} \Psi_j \underbrace{\begin{pmatrix} \Delta cr^r \\ \Delta e_P^r \\ \Delta e_O^r \\ \Delta p_A^r \\ \Delta \pi \\ \Delta i_L \\ \Delta i \end{pmatrix}_{t-i}}_{\text{Short-run dynamics}} + \Gamma s_t + \epsilon_t$$

Output - Financial Cycle Nexus

Effect on:	**Δcredit$_t$**	**Δ(consumption +investment)$_t$**	**Δasset prices$_t$**
from steady-state deviations			
Leverage gap$_{t-1}$	*Negative*	–	–
DSR gap$_{t-1}$	*Negative*	*Negative*	*Negative*
from short-run dynamics			
Δcredit$_{t-k}$	*Positive*	*Positive*	*Positive*

- Two gaps interact…
  - $\widetilde{lev}_t < 0$ implies high credit growth which supports output…
  - …but also implies higher debt-service…
  - $\widetilde{dsr}_t > 0$ drags on growth as well as credit
- …leading to endogenous cycles
  - Output effects large and very persistent
  - Crisis not result of shocks

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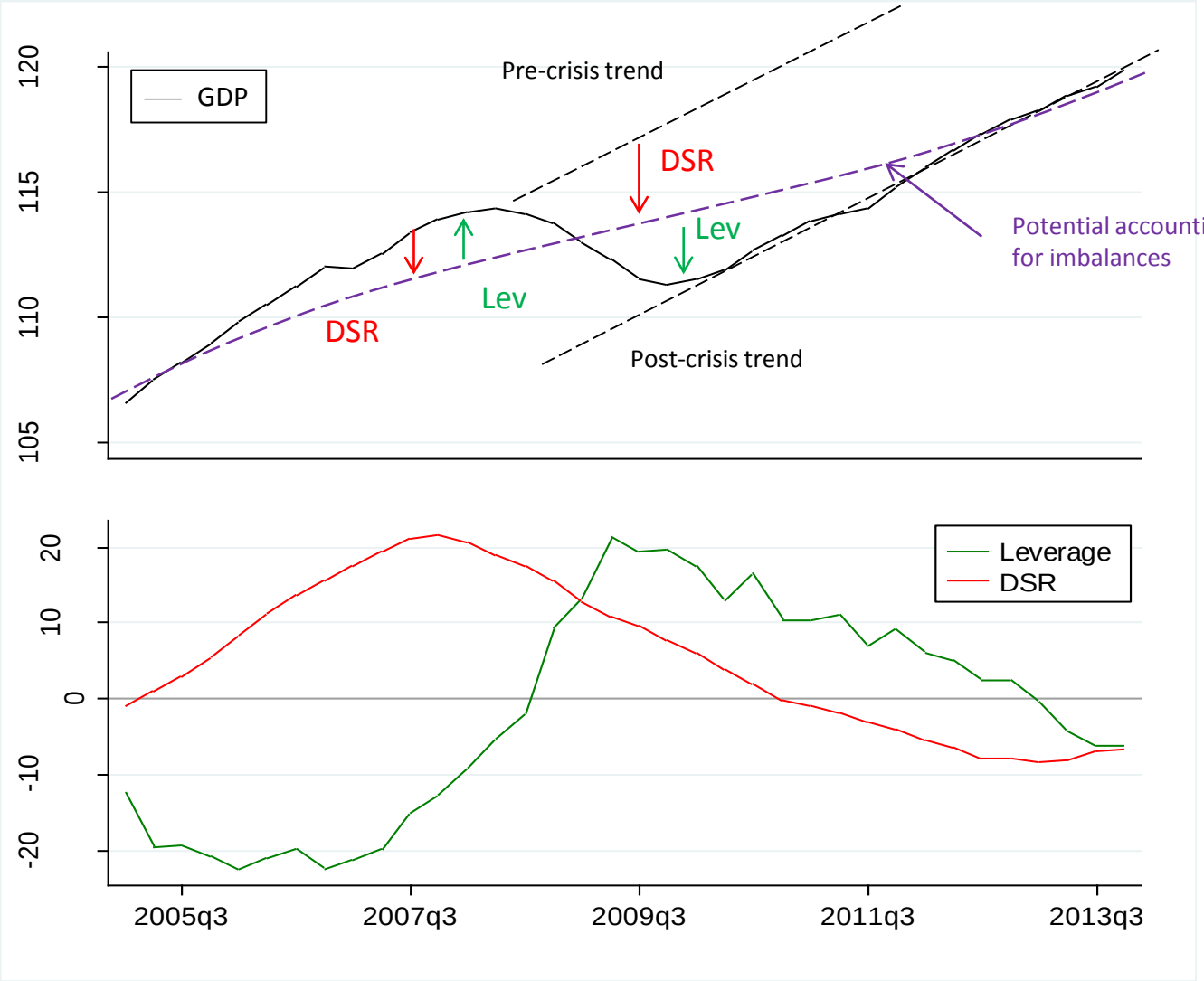
Output - Financial Cycle Nexus

Financial gaps and potential output



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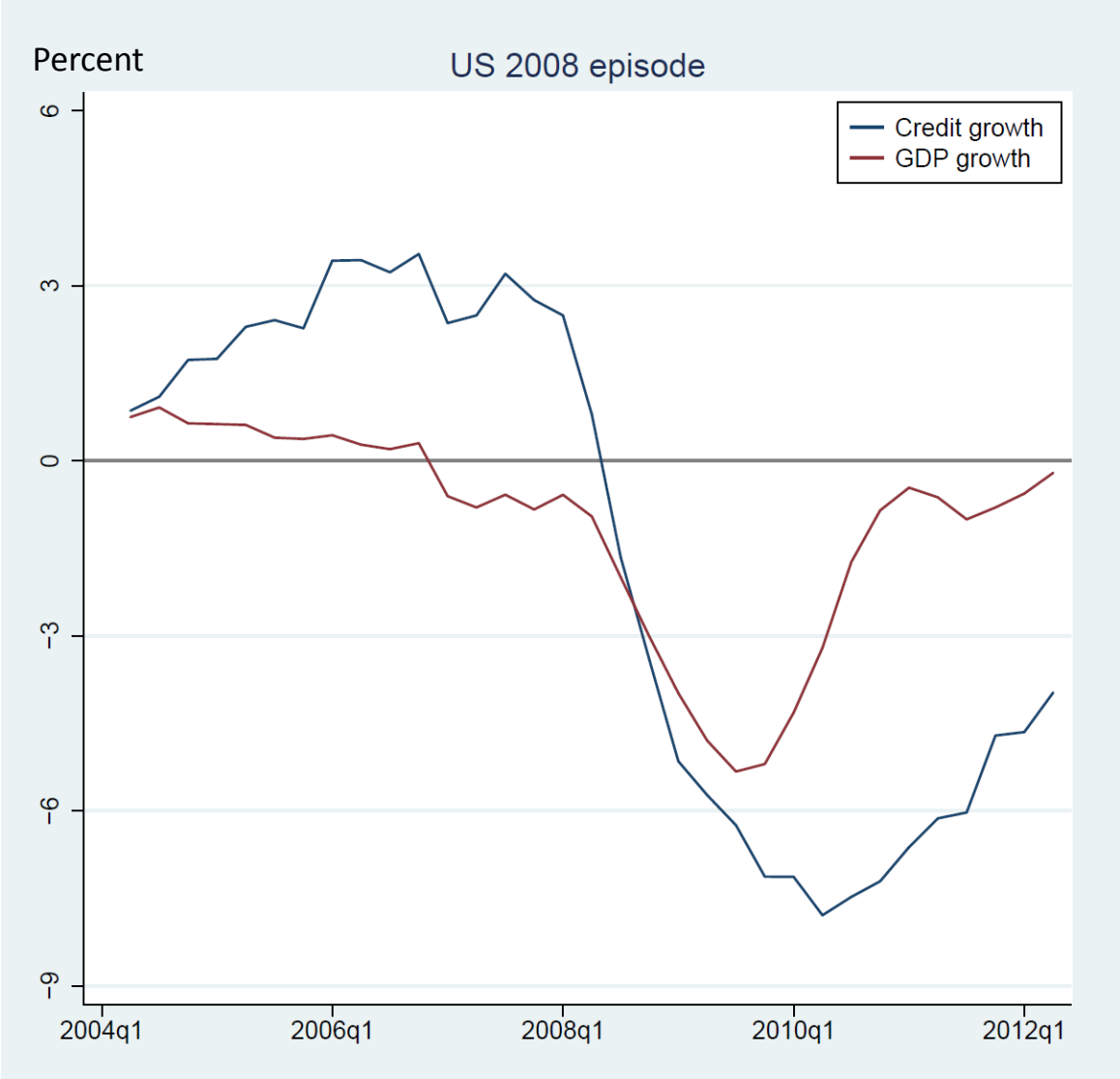
Output - Financial Cycle Nexus

Growth-less credit boom and credit-less recoveries



Trend Output and Interet Rate

Augmented LW filter

$$y_t - y_t^* = \beta_1(y_{t-1} - y_{t-1}^*) - \varphi_1(r_t - r_t^*) - \varphi_2 \widetilde{lev}_t + \vartheta_{1t}$$

$$\Delta y_t^* = \Delta y_{t-1}^* + \vartheta_{2t}$$

$$(\pi_t - \pi^*) = \beta_2(\pi_{t-1} - \pi^*) + \varphi_3(y_t - y_t^*) + \vartheta_{3t}$$

$$r_t^* = \beta_3 r_{t-1}^* + (1 - \beta_3)(z_t + \frac{1}{\rho} 4\Delta y_t^*) + \vartheta_{4t}$$

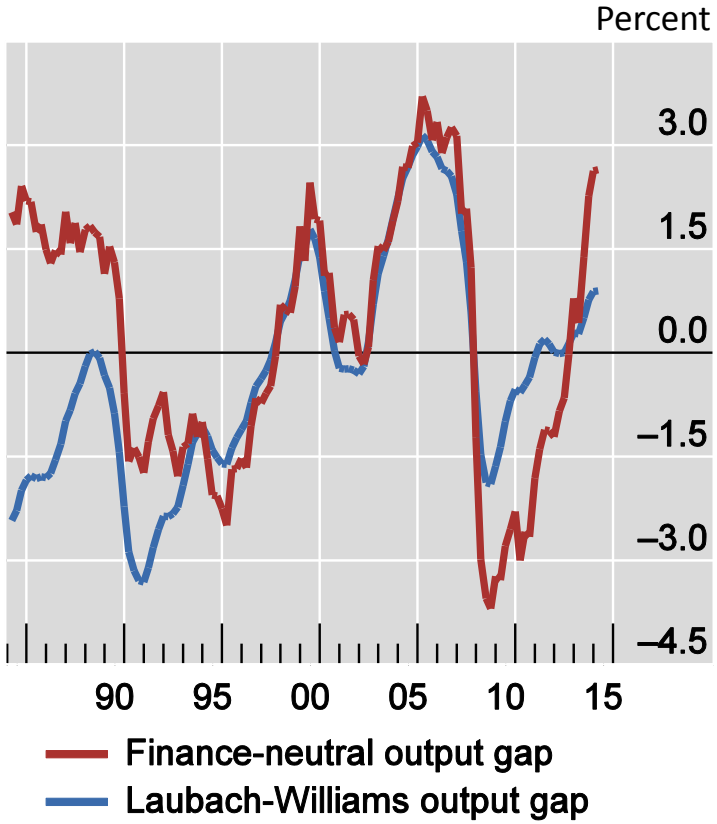
$$z_t = \beta_4 z_{t-1} + \vartheta_{5t}$$

$$\widetilde{lev}_t = \beta_5 \widetilde{lev}_{t-1} + \varphi_4(r_t - r_t^*) + \varphi_5 \widetilde{dsr}_{t-1} + \vartheta_{6t}$$

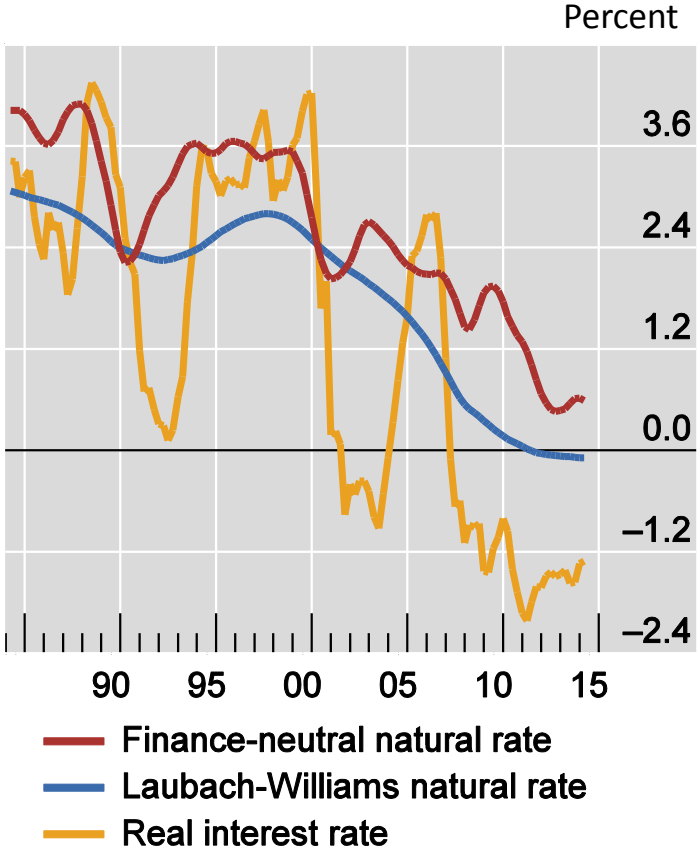
Finance-Neutral Output Gap and Natural Rate

Financial cycle affects trend output and interest rate estimates

Output Gap



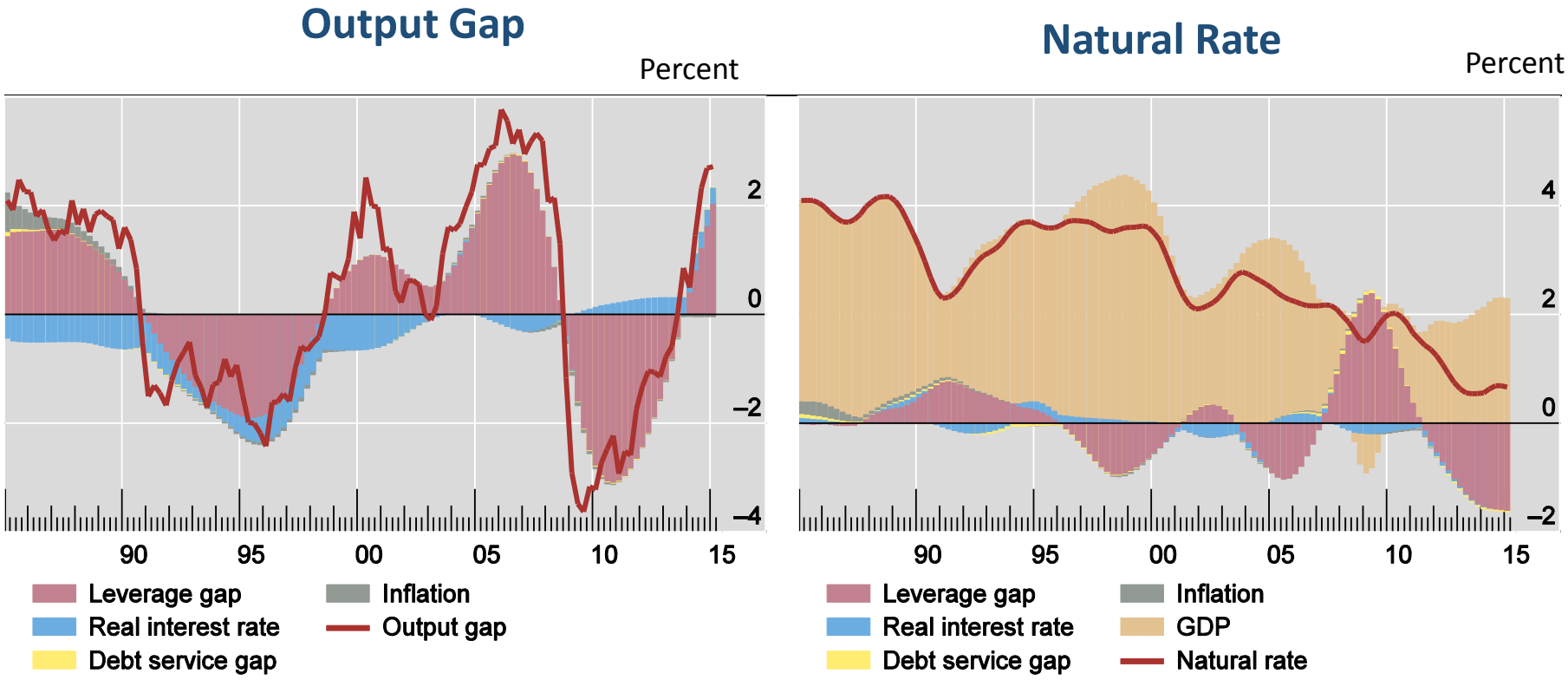
Natural Rate



Source: Juselius et al. (2016)

Finance-Neutral Output Gap and Natural Rate

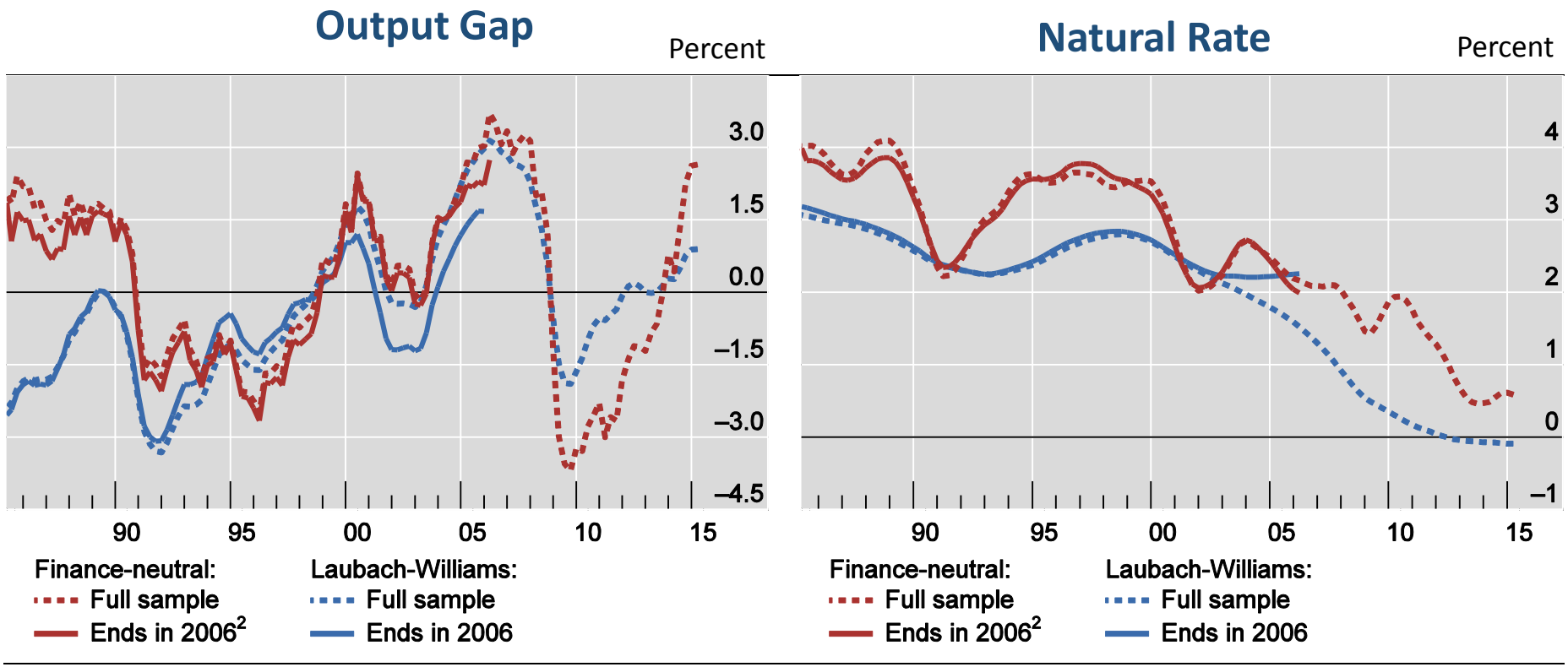
Variance Decompositions: Contribution of financial information



Source: Juselius et al. (2016)

Finance-Neutral Output Gap and Natural Rate

Finance-neutral gaps much more stable in real time



Source: Juselius et al. (2016)

What If…A Counterfactual

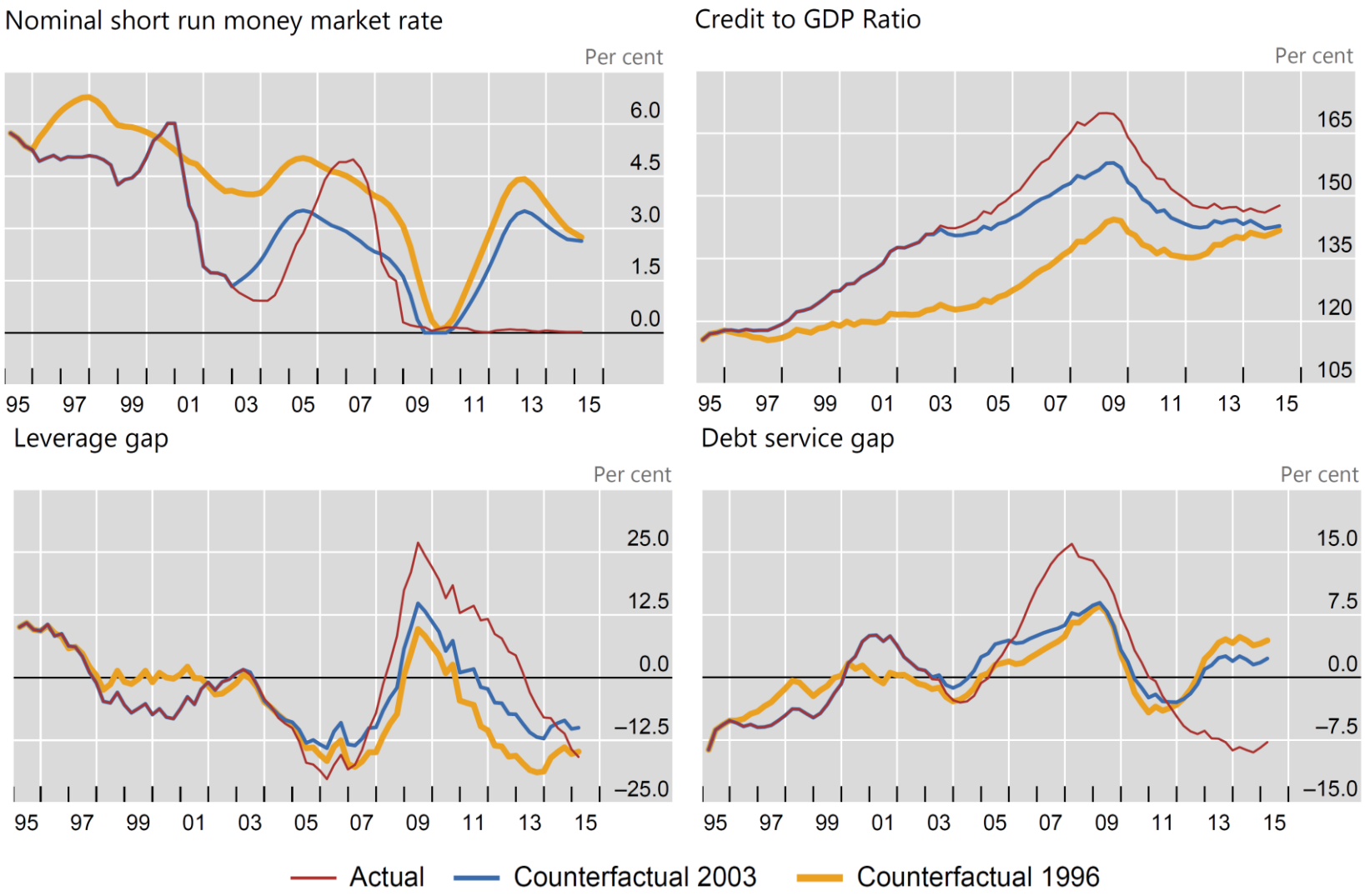
Alternative policy rule

$$i_t = \rho i_{t-1} + (1-\rho)\left(r^*_{t-1} + \pi^* + 1.5(\pi_{t-1} - \pi^*) + 0.5(y_{t-1} - y^*_{t-1}) - \lambda\, \widetilde{dsr}_{t-1}\right)$$

- Shocks the same – only policy rates change
  - Start in 2003 and 1996

Effects of monetary policy very persistent

Counterfactual Simulations



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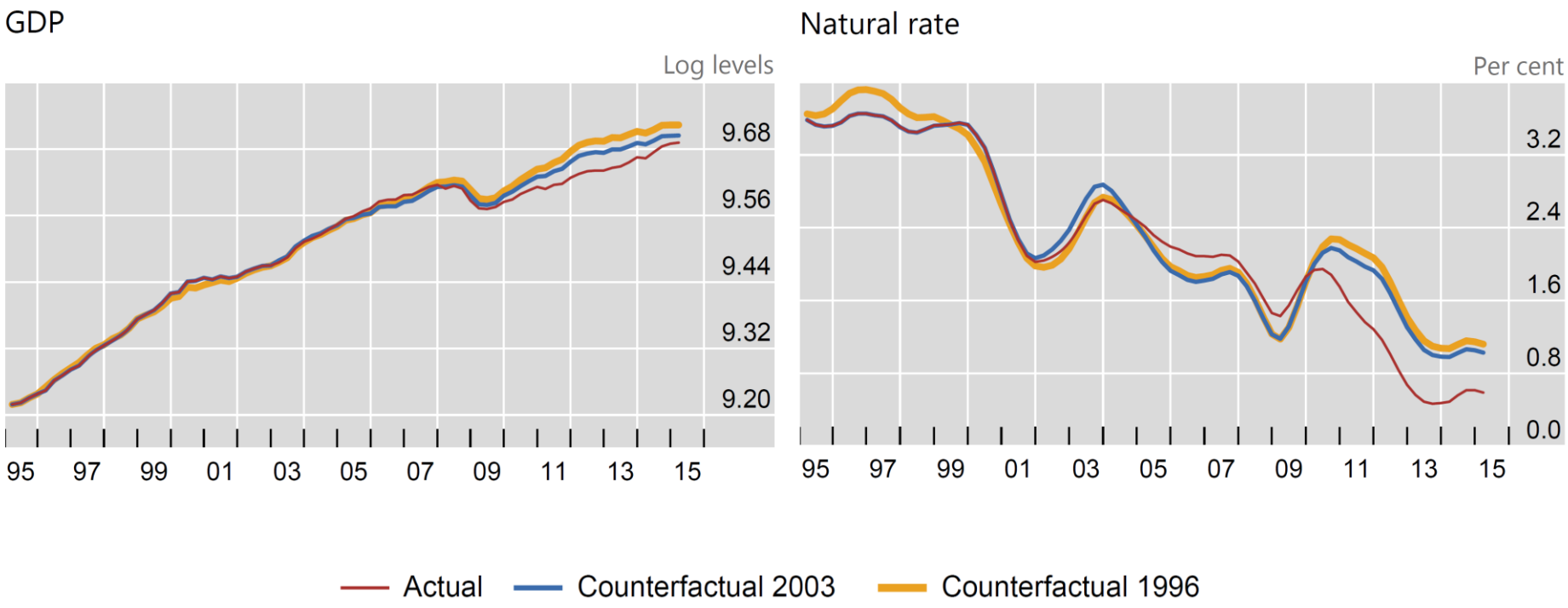
Effects of monetary policy very persistent

Counterfactual Simulations



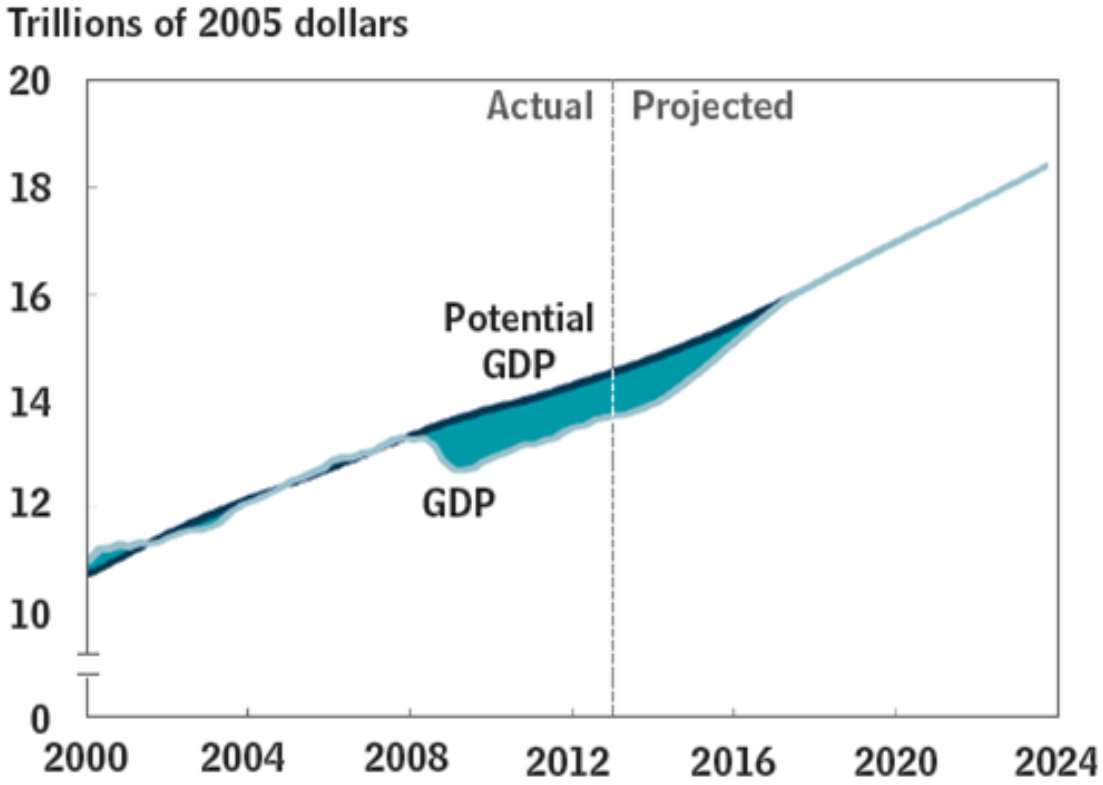
Conclusion

1. Financial cycle systematically affects real economy

- Financial cycle information helps in assessing trends
  - Better "sustainable" output gaps and real time stability
  - Natural rate higher and has fallen by less
- Cycle is endogenous to policy regimes and its impact very persistent
  - No clean separation of trend & cycle/short vs long run

Conventional views crisis as temporary shocks…

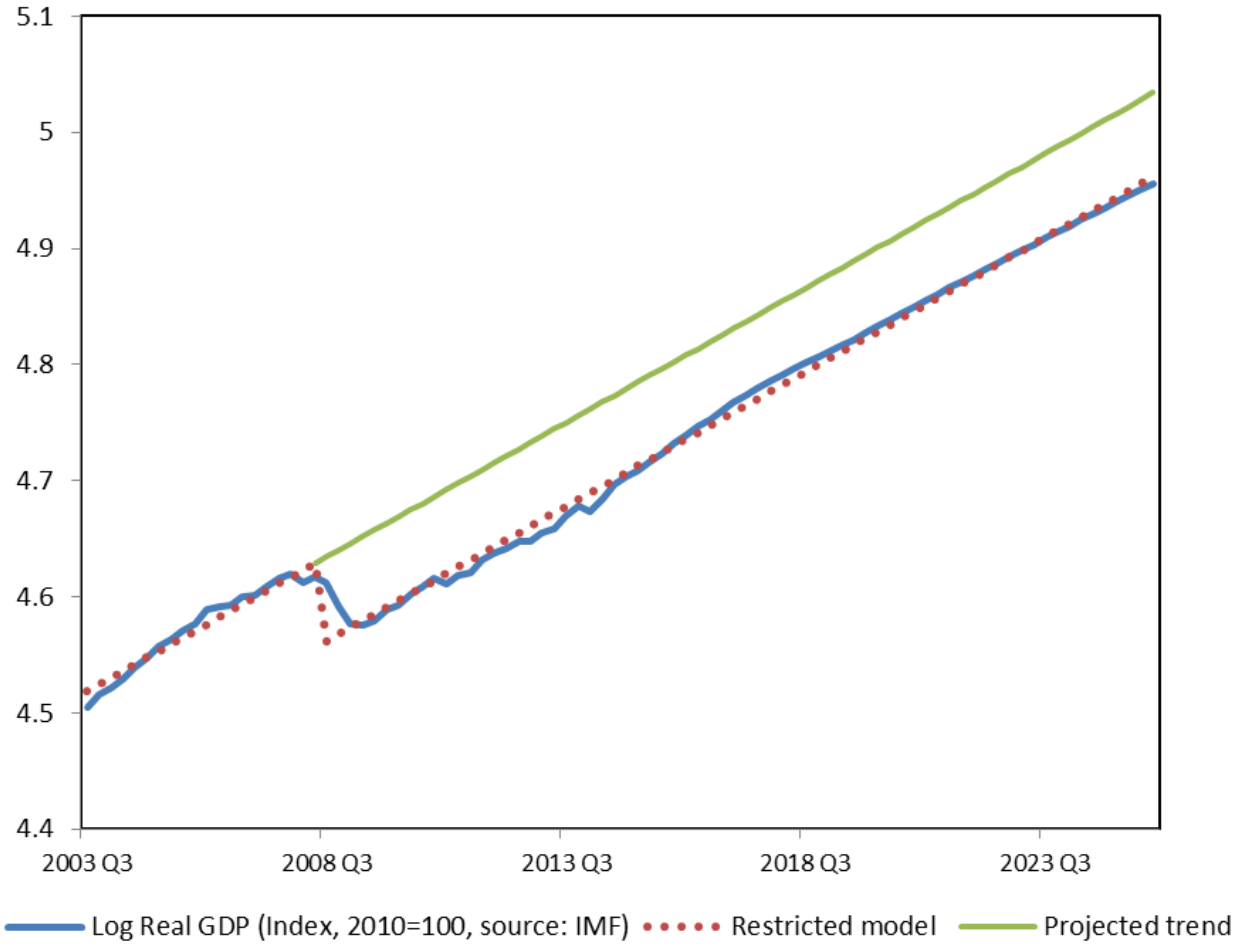
CBO Projections of US GDP



GDP and Potential GDP

Source: http://cbo.gov/sites/default/files/cbofiles/attachments/43907-BudgetOutlook.pdf

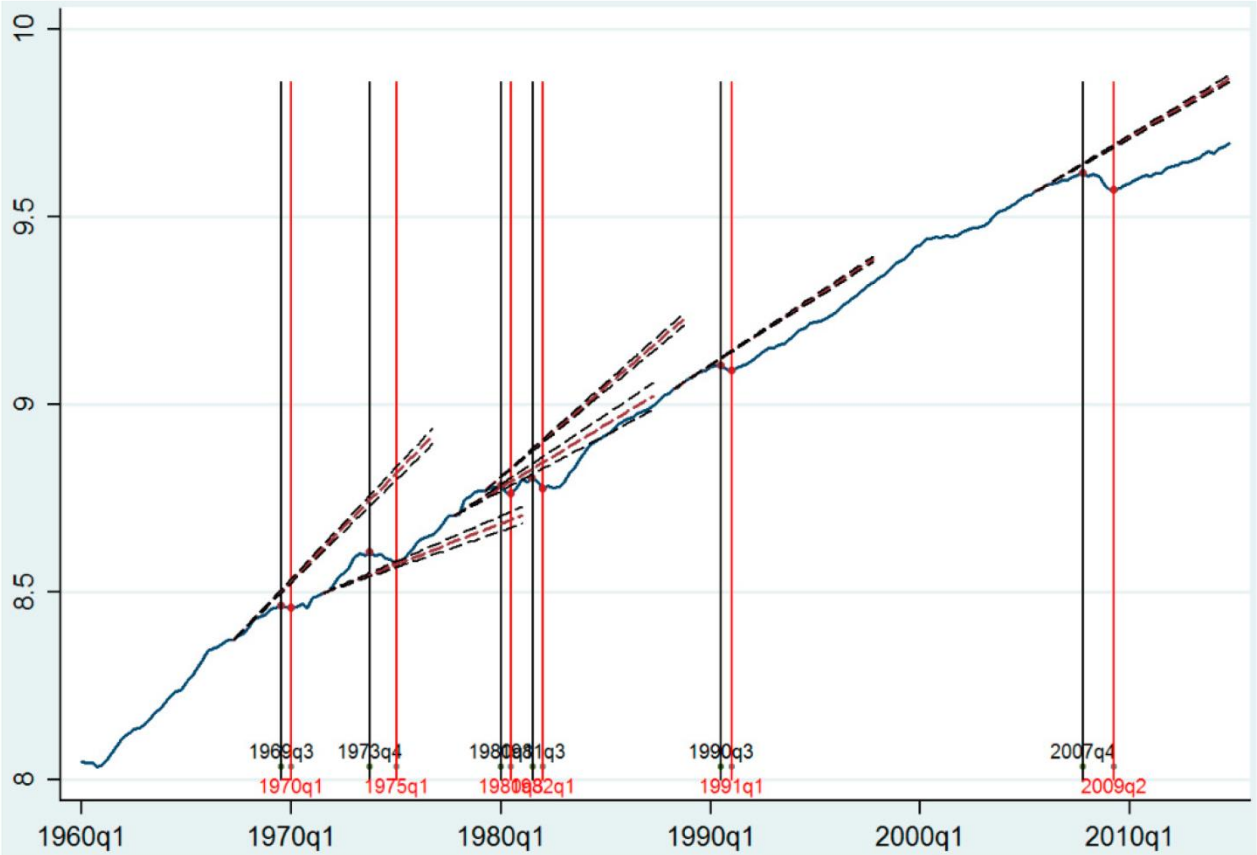
…but what if they have long-lasting impact?



Source: http://econbrowser.com/archives/2015/02/guest-contribution-long-term-effects-of-the-great-recession

…which seems to be the case

Figure 3: United States –Evolution of log real GDP and Extrapolated Trends



Source: Blanchard O, E Cerutti and L Summers (2015): "Inflation and Activity – Two Explorations and their Monetary Policy Implications," IMF Working Paper, No. 15/230.

Conclusion

2. What matters is *systematic* policy

- "Lean against the wind" often aim to avoid crisis by prompt action, but…
  - Output costs even without crisis
  - Path dependency implies that the *cumulative* role of policy matters
- …better to systematically keep finance on an even keel
  - Focus on <u>stocks</u> and <u>states</u> rather than <u>flows</u> and <u>shocks</u>
  - No trade-off in long run